

Application of NPP-VIIRS Nighttime Light Imagery for GDP Spatialization and Economic Differences in Vietnam

This project aims to investigate how NPP-VIIRS nighttime light imagery can be used to spatialize economic activity and identify regional economic differences in Vietnam. The main question is whether nighttime light intensity can serve as a useful proxy for GDP and provide a more detailed representation of economic activity across different regions. The project will use nighttime light imagery from the Earth Observation Group's VIIRS nighttime light dataset, available from the [Earth Observation Group](#), along with regional economic data, such as GDP by province or region. While GDP provides an important measure of economic activity, aggregated GDP statistics can obscure differences between regions and provide limited information about the geographic distribution of economic production. By analyzing the spatial distribution of nighttime lights and comparing it with regional GDP, this project will evaluate whether nighttime imagery can reveal patterns of economic activity that traditional GDP statistics may overlook (Chen & Nordhaus, 2011). The results could also help identify geographic factors associated with regional economic differences, such as rivers, elevation, transportation networks, and urbanization. More broadly, this research could demonstrate how remotely sensed data can complement traditional economic statistics and provide researchers and policymakers with a more detailed understanding of regional economic development in Vietnam.

References:

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Analyzing the Determinants of Pokémon Competitive Strength Through Data-Driven Segmentation

This project aims to investigate the factors that determine whether a Pokémon is considered strong or weak in competitive play. The main question is: after decomposing each Pokémon into different aspects, such as its base stats, typing, abilities, moves, and synergy, how important is each factor in determining its overall competitive strength, represented by its competitive usage? To answer this question, data will be collected from the Smogon Pokémon Showdown statistics page, particularly the 2026 statistics dataset on the new Pokemon Champion game, which provides information about Pokémon usage and competitive performance. The project will apply statistical and data analysis techniques to segment Pokémon based on their characteristics and examine which features are most strongly associated with competitive success. This analysis could provide a new perspective on Pokémon strength by moving beyond individual rankings and identifying the underlying characteristics that contribute to a Pokémon's performance. Understanding the relative importance of these factors could also provide useful insights for game balance. For example, the results could help identify which characteristics make certain Pokémon disproportionately strong and potentially inform the design or balancing of future Pokémon. More broadly, this project demonstrates how data-driven segmentation can be applied to understand complex systems in which multiple characteristics contribute to an overall outcome (Tenenbaum, 2024).

References:

<https://hdl.handle.net/1721.1/153888>

https://www.researchgate.net/publication/392469085_Predicting_Game_Character_Win_Probabilities_Using_Ensemble_Machine_Learning

